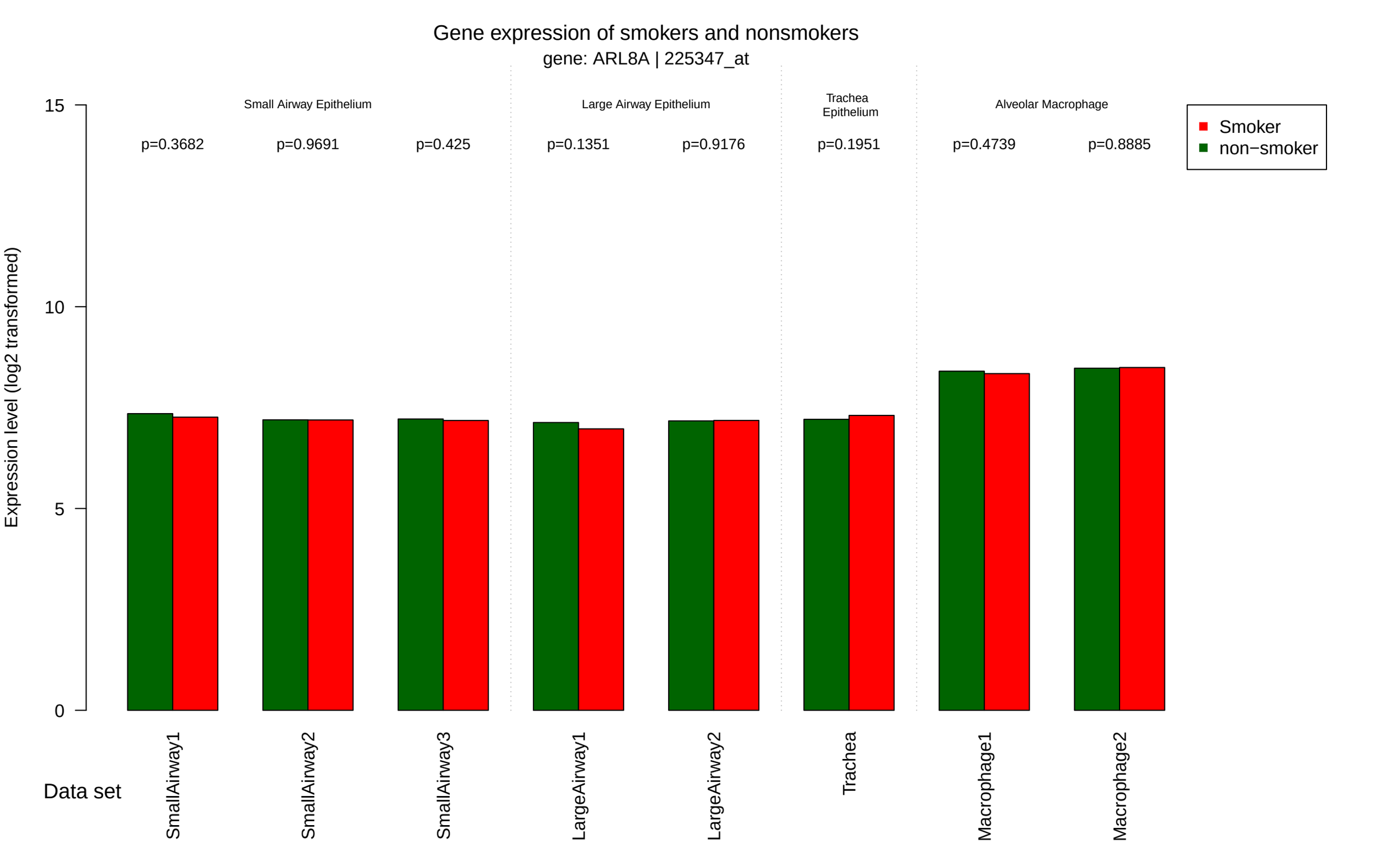
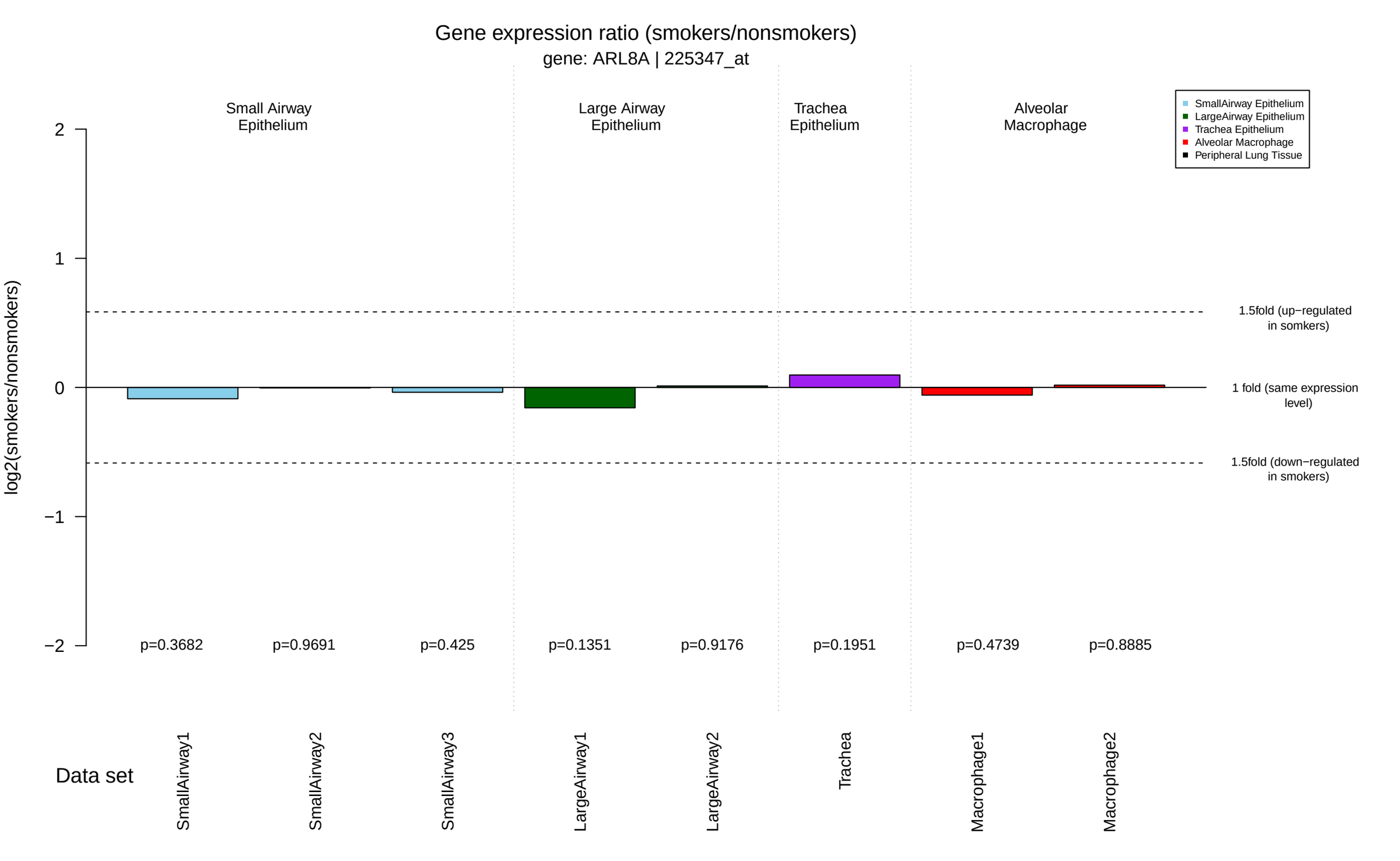
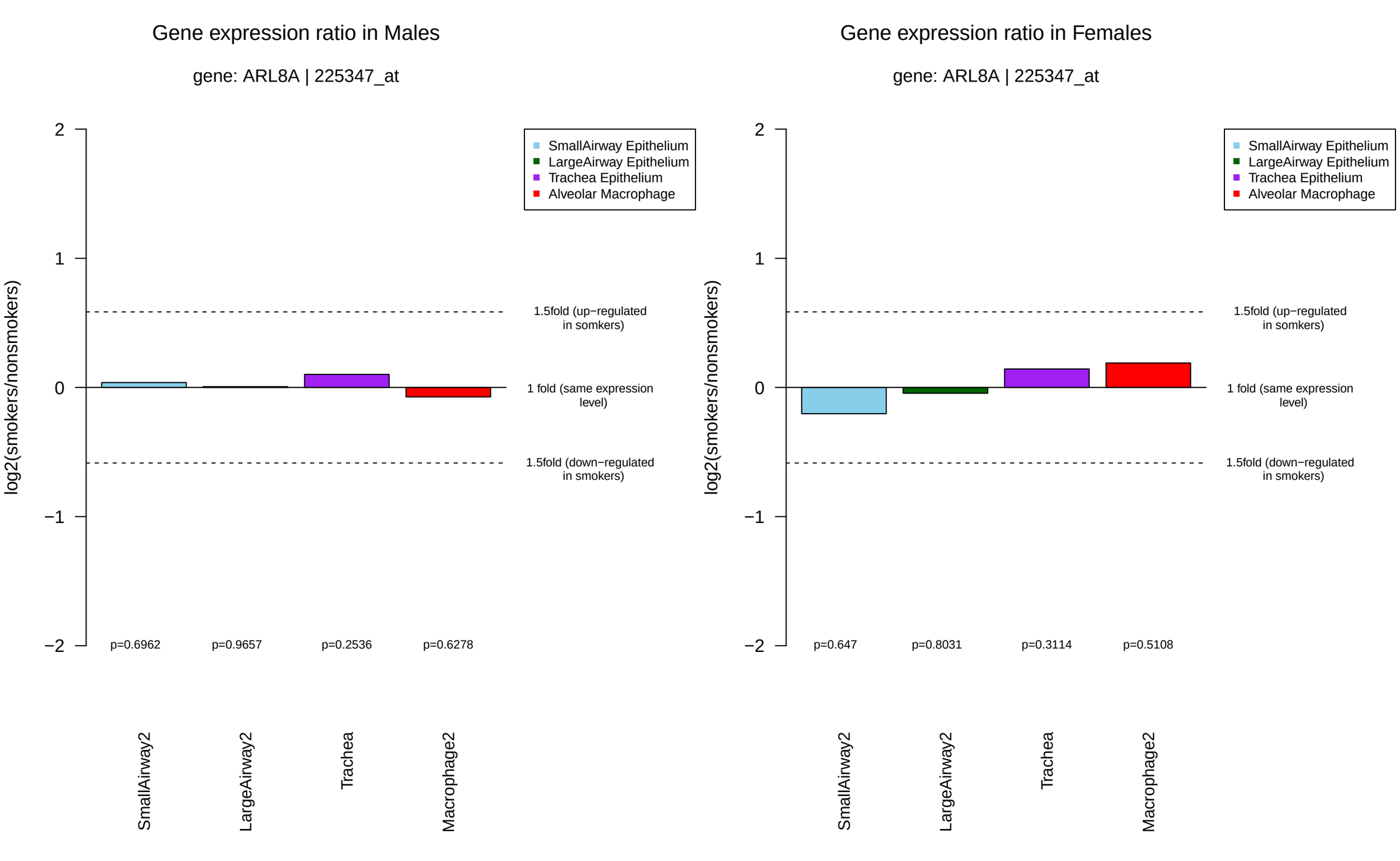
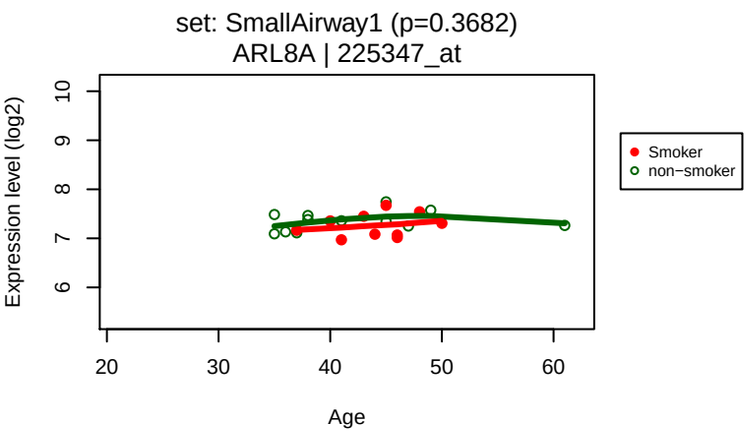
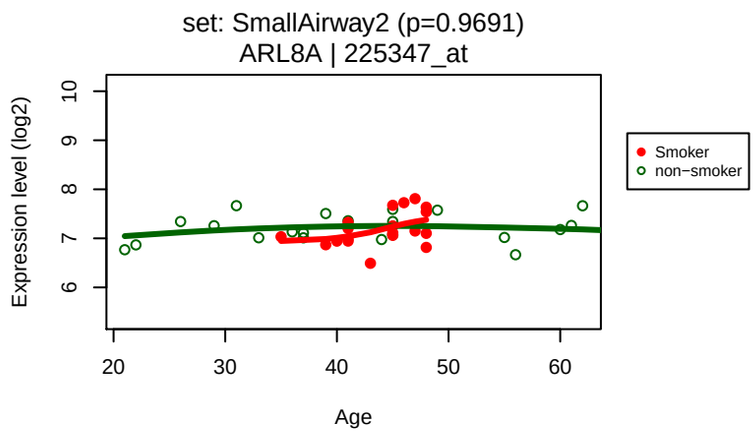
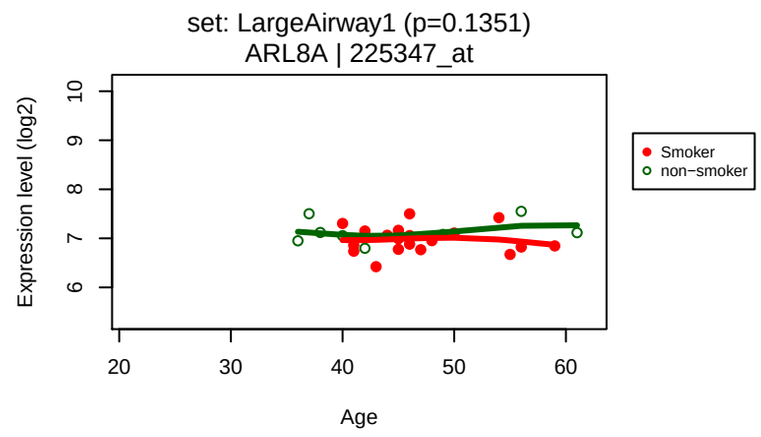

Gene expression of smokers and nonsmokers
gene: ARL8A | 225347_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3682
p=0.9691
p=0.425
p=0.1351
p=0.9176
p=0.1951
p=0.4739
p=0.8885
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

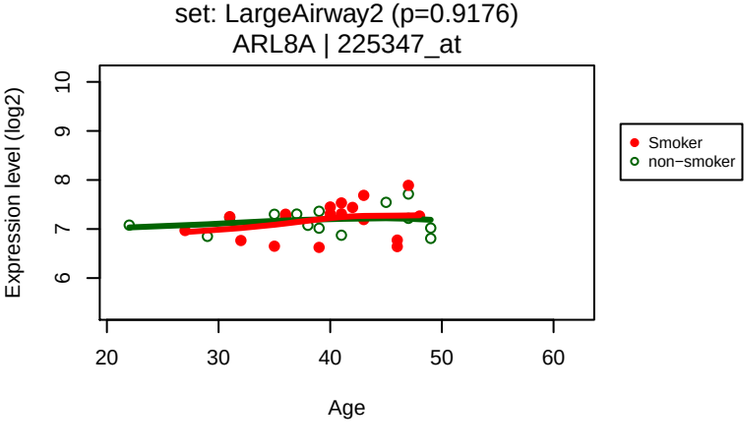

Gene expression ratio (smokers/nonsmokers)
gene: ARL8A | 225347_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3682
p=0.9691
p=0.425
p=0.1351
p=0.9176
p=0.1951
p=0.4739
p=0.8885
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

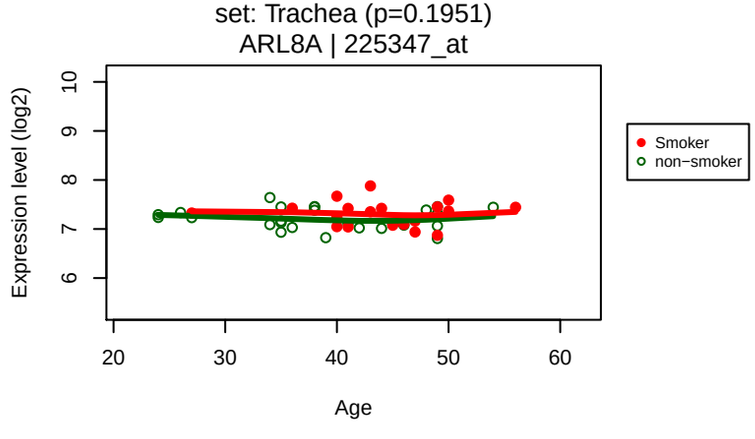

Gene expression ratio in Males
gene: ARL8A | 225347_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6962
p=0.9657
p=0.2536
p=0.6278
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARL8A | 225347_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.647
p=0.8031
p=0.3114
p=0.5108
SmallAirway2
LargeAirway2
Trachea
Macrophage2

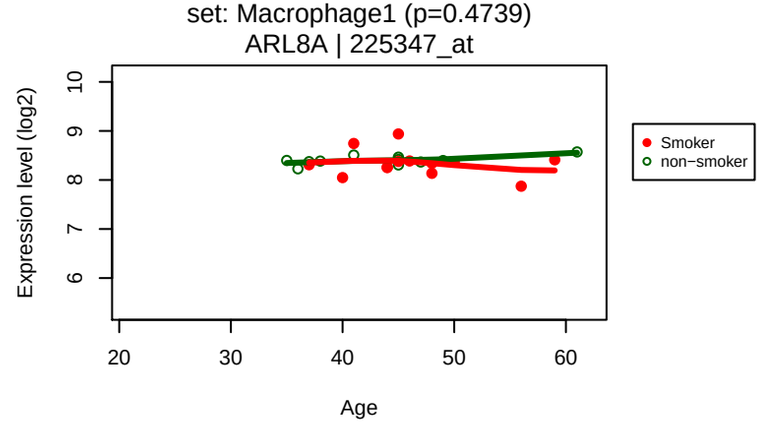

set: SmallAirway1 (p=0.3682)
ARL8A | 225347_at
Expression level (log2)
Age
Smoker
non−smoker

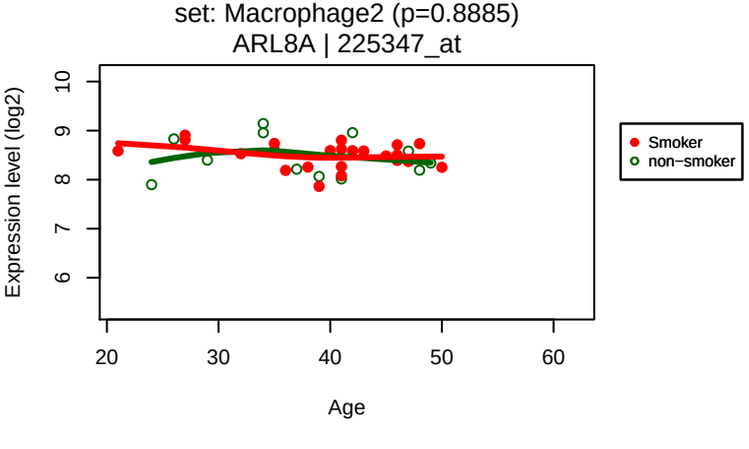

set: SmallAirway2 (p=0.9691)
ARL8A | 225347_at
Expression level (log2)
Age
Smoker
non−smoker

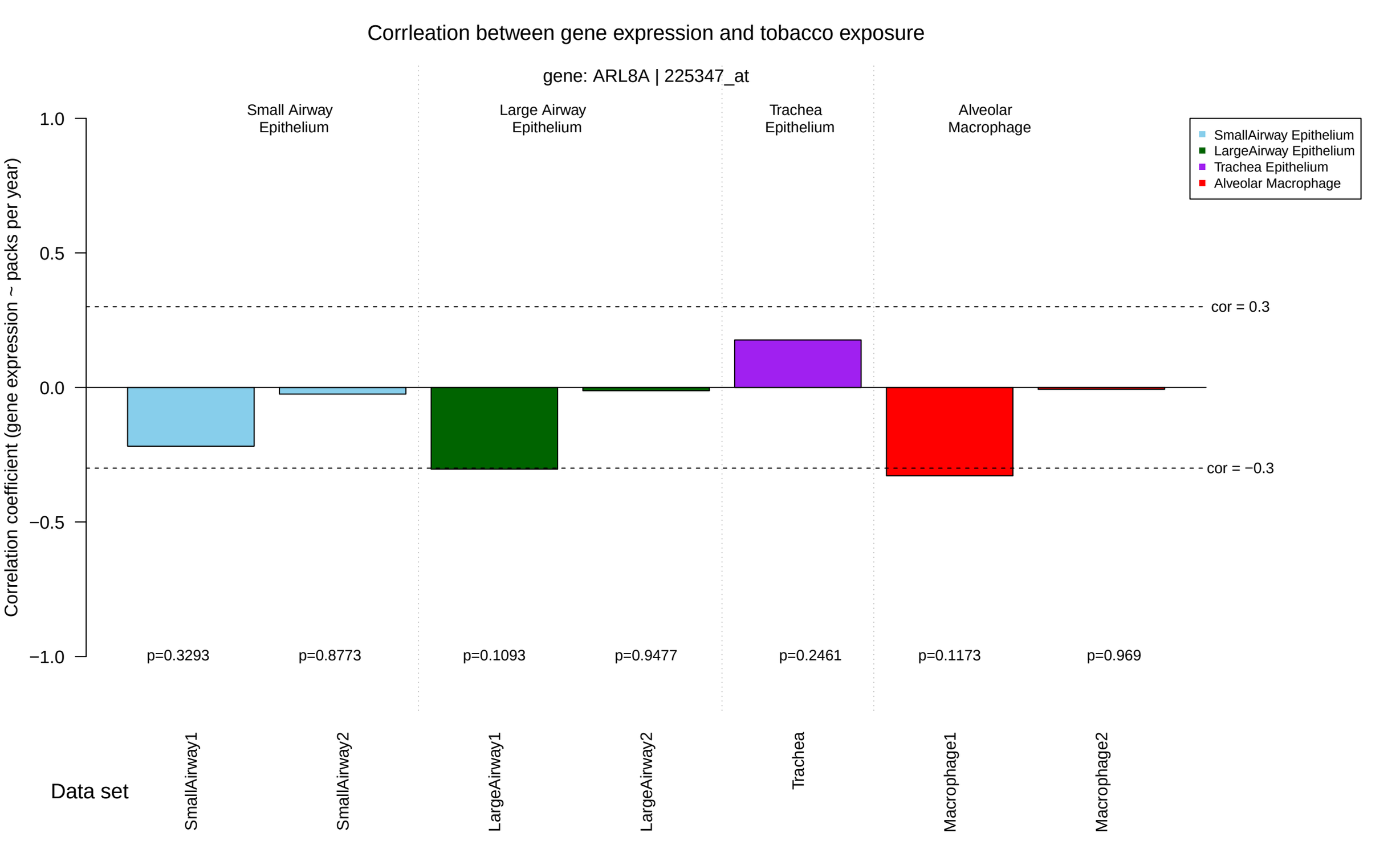

set: LargeAirway1 (p=0.1351)
ARL8A | 225347_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9176)
ARL8A | 225347_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1951)
ARL8A | 225347_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4739)
ARL8A | 225347_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8885)
ARL8A | 225347_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARL8A | 225347_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3293
p=0.8773
p=0.1093
p=0.9477
p=0.2461
p=0.1173
p=0.969
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2